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Medical device and methods of use

Abstract

The present disclosure is directed to a medical device. Systems and methods are provided for utilizing a laser to break a kidney stones into smaller fragments and/or dust, and removing particles, stone fragments and/or stone dust from a patient. The medical device may include a delivery device including a tube, and an elongate member having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end, wherein the elongate member is configured to move axially relative to the tube and apply suction through the distal end.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application is a continuation of U.S. application Ser. No. 16/679,406, filed Nov. 11, 2019, which is a divisional of U.S. application Ser. No. 15/415,604, filed Jan. 25, 2017, now U.S. Pat. No. 10,499,936, which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 62/288,734, filed Jan. 29, 2016, each of which is herein incorporated by reference in its entirety.

TECHNICAL FIELD
(1) The present disclosure relates generally to medical devices. More particularly, the disclosure relates to medical devices used, for example, in breaking objects into smaller particles, and removing the resulting particles from a patient. The disclosure also relates to methods of using such devices.
BACKGROUND
(2) The incidence of hospitalization for the removal of urinary calculi, commonly referred to as kidney stones, has been estimated at 200,000 cases per year. A vast majority of these patients pass their stones spontaneously; however, in the remainder, the kidney stone(s) become impacted in the ureter, a muscle tube joining the kidney to the bladder. An impacted kidney stone is a source of intense pain and bleeding, a source of infection and, if a stone completely blocks the flow of urine for any extended length of time, can cause the loss of a kidney.
(3) Recently, various methods have been utilized to break the stone into smaller fragments. One such method is stone dusting. Stone dusting is used by some urologists to fragment and evacuate stones from a kidney and is often performed by a ureteroscope. Intense light energy from a laser within the ureteroscope breaks the stone into increasingly smaller pieces. However, in some cases, the stone and/or the stone fragments may be pushed away from the ureteroscope by the laser, thus making it impossible to continue to break the stone or stone fragments into smaller fragments without repositioning the ureteroscope. The disclosure addresses the above-mentioned process and other problems in the art.
(4) Further, rather than breaking up the stone into chunks, which are removed by baskets, dusting generates very small fragments that are capable of being passed naturally. However, in some cases,

these small stone fragments may not pass naturally. For example, the stone fragments may collect in an area of the kidney where they are less likely to flow out naturally, such as the lower calyx of the kidney. In theory, any of these small stone fragments that do not evacuate through natural urine flow, could be a seed for new stone growth. Thus, the application of suction may be employed to remove the stone dust. Breaking up a stone and providing suction requires a working channel with a sufficient internal cross-sectional area to receive a laser fiber and a lumen with sufficient internal cross-sectional area to allow stone fragments and/or dust to pass through without clogging. The combined cross-sectional areas of these two elements may make a device too large to reach the target kidney stone. For example, the kidney stone may be within the kidney, or, specifically, within the lower calyx of the kidney. Often, the space within the kidney and/or lower calyx of the kidney is more limited than the space within the ureter and this space may not be large enough to accommodate both a working channel for a laser fiber and a lumen for applying suction. The disclosure addresses the above-mentioned process and other problems in the art.

SUMMARY OF THE DISCLOSURE

(5) Aspects of the present disclosure provide device and methods for breaking an object into smaller particles and removing said particles from portions of the human body.

(6) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive.

(7) In one example, a medical device may include a delivery device including a tube, and an elongate member having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end, wherein the elongate member is configured to move axially relative to the tube and apply suction through the distal end.

(8) Examples of the medical device may additionally and/or alternatively include one or more other features. Features of the various examples described in the following may be combined unless explicitly stated to the contrary. For example, the proximal end of the elongate member may be forked. The proximal end of the elongate member may be connected to a control system. The control system may control the vacuum source. The proximal end of the elongate member may be in fluid communication with a vacuum source. The medical device may include a sleeve, wherein the sleeve may be configured to receive the tube and the elongate member simultaneously. The delivery device may be one of ureteroscope, sheath, catheter, or endoscope. The medical device may include a laser fiber connected to the tube. The lumen of the elongate member may be substantially D-shaped. The tube may include an illumination device. The tube may include an imaging device. The tube may have a first engaging surface, and the elongate member may have a second engaging surface, and the first engaging surface may be substantially complementary of the second engaging surface. The elongate member may be in fluid communication with a fluid source. The lumen of the elongate member may have an inner diameter of approximately 1 mm to approximately 3.5 mm. The elongate member may be at least partially controlled by a control system and the tube may be at least partially controlled by a handle.

(9) In another example, a method operating a medical device may include positioning a distal end of an elongate member of the medical device at a target area, applying suction through a lumen of the elongate member to seal an object to the distal end of the elongate member, and moving the distal end of the elongate member to move the object sealed to the distal end of the elongate member.

(10) Examples of the method of operating the medical device may additionally and/or alternatively include one or more other features. For example, prior to positioning the distal end of the elongate member of the medical device at the target area, the method may include positioning the distal end of the elongate member proximal to the target area, and supplying dye through the lumen of the elongate member to the distal end of the elongate member. The method may include moving the object proximate to a tube, disposing a laser fiber within the tube, and initiating the laser.

(11) In another example, a method operating a medical device may include positioning a distal end

of an elongate member of the medical device distal to a distal end of a tube of the medical device, applying suction through a lumen of the elongate member to seal a kidney stone to the distal end of the elongate member, and moving the distal end of the elongate member to move the kidney stone sealed to the distal end of the elongate member proximally toward the distal end of the tube.
(12) Examples of the method of operating the medical device may additionally and/or alternatively include one or more other features. For example, the method may include disposing a laser fiber within the tube, and initiating the laser.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various examples and together with the description, serve to explain the principles of the disclosure.
(2) FIG. **1** illustrates an exemplary medical device, including a tube, an elongate member, a handle portion, a control system, and a sleeve;
(3) FIG. **2**A illustrates an exemplary distal end of the medical device of FIG. **1** with the elongate member in a retrieval position;
(4) FIG. **2**B illustrates an exemplary distal end of the medical device of FIG. **1** with the elongate member in an operating position;
(5) FIG. **3**A illustrates an exemplary cross-section of the medical device taken at **3-3** of FIG. **1**;
(6) FIG. **3**B illustrates an alternative exemplary cross-section of the medical device taken at **3-3** of FIG. **1**;
(7) FIG. **4** illustrates an exemplary connection between a proximal end of the elongate member and the control system of FIG. **1**;
(8) FIGS. **5**A-C illustrates an exemplary method of operation of the medical device of FIG. **1**; and
(9) FIG. **6** is a block diagram of an exemplary method of using the medical devices disclosed herein.
DETAILED DESCRIPTION
(10) Reference is now made in detail to examples of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. The term “distal” refers to a position farther away from a user end of the device. The term “proximal” refers a position closer to the user end of the device. As used herein, the terms “approximately” and “substantially” indicate a range of values within +/−5% of a stated value.
Overview
(11) Aspects of the present disclosure relate to systems and methods for breaking kidney stones into smaller particles and removing those particles from a target area of a patient's body. The medical device described herein may work by securing an elongate member for suction to an outer surface of a tube of a delivery device. A distal end of the tube and a distal end of the elongate member may be positioned within a ureter of a patient. In some examples, radiopaque dye may be injected into the target area, e.g., the kidney, from a lumen of the elongate member. Then, the elongate member may be extended distally to a retrieval position in which the distal end of the elongate member is positioned within the target area, e.g., the lower calyx of the kidney, and/or proximate the target kidney stone. Suction may be applied through the elongate member, pulling the stone toward a distal opening of the elongate member until the stone contacts and forms a seal with the distal opening of the elongate member. The formed seal secures the kidney stone to the distal opening of the elongate member. With the stone secured to the elongate member, the elongate member may be retracted or pulled proximally toward the distal end of the tube until the elongate member is in the

operating position, e.g., the secured stone is proximate a distal opening of the tube. At this point, a laser fiber, disposed within the tube, may be used to break up kidney stones into particles. It should be noted that the stone may be broken into particles in any way known in the art, including the application of ultrasound and/or sound waves on the stone. Once the stone is broken into particles that are smaller than the distal opening the elongate member, the seal may be broken and/or particles may be suctioned into the lumen of the elongate member. These particles may be vacuumed through the elongate member and out of the patient's body.

Detailed Examples

(12) FIG. **1** illustrates medical device **100** for removing stones from a patient's body. Medical device **100** may include a retrieval system. The retrieval system may include elongate member **202** and control system **210**. Elongate member **202** may be a hollow, flexible, elongate tube having a distal end **204** and a proximal end **206** and at least one lumen (e.g., lumen **214** of FIGS. **2**A-B) extending therebetween. Proximal end **206** of elongate member **202** may be coupled to control system **210**, vacuum source **250**, and/or fluid source **280**. As shown in FIG. **1**, medical device **100** may include a delivery device. The delivery device may be, for example, any traditional delivery device, including a ureteroscope, sheath, catheter, endoscope, etc. The delivery device may include a tube **102** and a handle portion **110**. Tube **102** may be a hollow, flexible, elongate tube having a distal end **104** and a proximal end **106** and at least one lumen (e.g., lumen **112** of FIGS. **2**A-B) extending therebetween. Proximal end **106** of tube **102** may be coupled to handle portion **110**. The handle portion **110** and/or the proximal end **106** of tube **102** may be attached to a laser control **130** and/or a fluid supply assembly **140**. In some examples, medical device **100** may include a sleeve **302** and fastener **136**. Sleeve **302** and fastener **136** may be any mechanism known in the art for allowing elongate member **202** (and/or the entirety of the retrieval device) to move axially relative to tube **102** and prevent radial movement of elongate member **202** relative to tube **102**.

(13) Depending upon the particular implementation and intended use, the length of tube **102**, elongate member **202**, and/or sleeve **302** may vary. In the example shown in FIG. **1**, sleeve **302** may be shorter in length than tube **102** and elongate member **202**. Depending upon the particular implementation and intended use, tube **102**, elongate member **202**, and/or sleeve **302** can be rigid along its entire length, flexible along a portion of its length, or configured for flexure at only certain specified locations. In one example, tube **102**, elongate member **202**, and/or sleeve **302** may be flexible, adapted for flexible steering within bodily lumens, as understood in the art. For example, tube **102** can include a steering system (not shown) to move at least a portion (e.g., distal end **104**) up/down and/or side-to-side. Additional degrees of freedom, provided for example via rotation, translational movement of tube **102**, or additional articulation of bending sections, may also be implemented. Examples of such steering systems may include at least one of or more of pulleys, control wires, gearing, and electrical actuators.

(14) Tube **102**, elongate member **202**, and/or sleeve **302** may be formed of any suitable material having sufficient flexibility to traverse body cavities and tracts. In general, tube **102**, elongate member **202**, and/or sleeve **302** may be made of any suitable material that is compatible with living tissue or a living system. That is, tube **102**, elongate member **202**, and/or sleeve **302** may be non-toxic or non-injurious, and it should not cause immunological reaction or rejection. In some examples, tube **102**, elongate member **202**, and/or sleeve **302** may be made of polymeric elastomers, rubber tubing, and/or medically approved polyvinylchloride tubing. Polymeric elastomers may be, for example, EVA (Ethylene vinyl acetate), silicone, polyurethane, and/or C-Flex.

(15) Further, tube **102**, elongate member **202**, and/or sleeve **302** may include any suitable coating and/or covering. For example, the outer surface may include a layer of lubricous material to facilitate insertion through a body lumen or through sleeve **302**. Further, tube **102**, elongate member **202**, and/or sleeve **302** may be coated with a biocompatible material such as Teflon. To inhibit bacterial growth in the body cavity, tube **102**, elongate member **202**, and/or sleeve **302** may

be coated with an antibacterial coating. Further, an anti-inflammatory substance may also be applied to the outer surface of the tube **102**, elongate member **202**, and/or sleeve **302**, if required.
(16) Tube **102**, elongate member **202**, and/or sleeve **302** may be designed to impose minimum risk to the surrounding tissues while in use. To this end, one or more portions of tube **102**, elongate member **202**, and/or sleeve **302** may include atraumatic geometrical structures, such as rounded or beveled terminal ends or faces (e.g., distal end **104** of tube **102**), to reduce trauma and irritation to surrounding tissues.
(17) I. Sleeve
(18) Sleeve **302** may be hollow and configured to slidably receive tube **102** and elongate member **202** simultaneously. Tube **102** may be circular, ovoidal, irregular, and/or any shape suitable to enter a body and/or receive tube **102** and elongate member **202**. In some examples, a lumen running through sleeve **302** may be shaped to accommodate the size and shape of tube **102** and elongate member **202**. In other examples, the lumen may be substantially circular and sleeve **302** may be made of elastic material to stretch to accommodate the size and shape of the combination of tube **102** and elongate member **202**. A lumen extending through sleeve **302** may include any suitable coating. For example, the lumen may include a layer of lubricous material, for example, to facilitate insertion of tube **102** and elongate member **202**.
(19) II. Delivery Device
(20) As previously described, medical device **100** may include a delivery device. The delivery device may be, for example, any delivery device known in the art, including ureteroscope, sheath, catheter, endoscope, etc. The delivery device may include a tube **102** and a handle portion **110**.
(21) A. Handle Portion
(22) Handle portion **110** can be attached to proximal end **106** of tube **102** by, for example, welding, a locking configuration, use of an adhesive, or integrally forming with tube **102**. The handle portion **110** may include a plurality of ports. For example, a first port may supply to distal end **104** of tube **102** a laser fiber (e.g., laser fiber **120**) coupled to laser control (e.g., laser control **130**). In some examples, a second port may place lumen **112** in fluid communication with a fluid supply assembly (e.g., fluid supply assembly **140**). Additional ports and lumens may be provided for supplying and/or controlling an illumination device and/or an imaging device located at or near distal end **104** of tube **102**. The handle portion **110** may include an actuating mechanism (not shown) to actuate one or more medical devices that may be located at the distal end **104** of tube **102**. For example, handle portion **110** may include an actuating mechanism to power on or off the laser, the illumination device, and/or the imaging device.
(23) B. The Tube
(24) Tube **102** may be circular, ovoidal, irregular, and/or any shape suitable to enter a body. Further, tube **102** may have a uniform shape along its length, or may having a varying shape, such as a taper at the distal end to facilitate insertion within the body. In some embodiments of the present disclosure, the distal end **104** of tube **102** includes visualization devices such as a camera and/or a light source. These devices may attach to the distal end **104** using known coupling mechanisms. Alternatively, the visualization devices may be detachably introduced into tube **102** through lumen **112** (or a separate lumen, e.g., lumen **124** or **126** of FIGS. **2**A-B) when required.
(25) Lumen **112** may be defined by an elongate hollow lumen that extends between proximal end **106** and distal end **104** of tube **102**. Lumen **112** may be any size, shape, and/or in any configuration within tube **102**. Exemplary cross-sections of tube **102** and lumen **112** will be described in further detail below with respect to FIGS. **3**A and **3**B.
(26) III. Retrieval System
(27) As previously described, medical device **100** may include a retrieval system. The retrieval system may include elongate member **202** and control system **210**.
(28) A. The Elongate Member
(29) Elongate member **202** may be circular, ovoidal, irregular, and/or any shape suitable to enter a

body and/or sleeve **302**. Further, elongate member **202** may have a uniform shape along its length, or may having a varying shape, such as a taper at the distal end to facilitate insertion within the body. In some embodiments of the present disclosure, the distal end **204** of elongate member **202** includes reinforced section **226**. The walls of reinforced section **226** may be thicker than the remainder of elongate member **202**. This may prevent distortion of the reinforced section **226** when a stone is vacuumed and held against a distal opening/distal end **204** (as described in further detail below with respect to FIGS. **5**A-C).

(30) Lumen **214** may be defined by an elongate hollow lumen that extends between proximal end **206** and distal end **204** of elongate member **202**. Lumen **214** may be any size, shape, and/or in any configuration within elongate member **202**. For example, the inner diameter of lumen **214** may be approximately 1 mm to approximately 3.5 mm, approximately 1.5 mm to approximately 3 mm, or approximately 2 mm to approximately 3 mm. Exemplary cross-sections of elongate member **202** and lumen **214** will be described in further detail below with respect to FIGS. **3**A and **3**B.

(31) In some examples, elongate member **202** may act as a guidewire to guide attached tube **202**. For example, elongate member **202** may be inserted into the body before the insertion of tube **102**, and tube **102** may be tracked along an outer surface (e.g., engaging surface **284**) of elongate member **202** in order to position tube **102** within the body. It should be noted that any of lumens **112** or **214** may additionally or alternatively be configured to receive a guidewire. In some embodiments, lumen **214** may receive a guidewire to facilitate positioning elongate member **202** in the body.

(32) To effectively maneuver the elongate member **202** within a body cavity and, for example, position distal end **204** proximate of a kidney stone within a lower calyx of a kidney, an operator may need to know the exact location of the elongate member **202** in the body. To this end, one or more portions of elongate member **202** may be made of radiopaque, such as by inclusion of barium sulfate in plastic material or inclusion of one or more metal portions, which provide sufficient radiopacity. In some examples, elongate member **202** may have a radiopaque coating. Additionally or alternatively, distal end **204** of elongate member **202** may include radiopaque or sonoreflective markers (not shown). These markings facilitate detection of a position and/or orientation of elongate member **202** within a patient's body, and an operator, with the aid of suitable imaging equipment, may track the path followed by elongate member **202**. This may help the operator avoid potential damage to sensitive tissues. By using fluoroscopic guidance, elongate member **202** may be located without the visualization devices within tube **102** and, therefore, move independently of and/or not within the field of view of these visualization devices. Further, with such guidance, the space within elongate member **202** that would be needed for direct visualization (e.g., an imaging apparatus) may instead be used to maximize the size of the lumen and/or the flow rate of the suction.

(33) FIGS. **2**A-B illustrate exemplary distal ends of medical device **100** with elongate member **202** in a retrieval position and an operating position, respectively. As shown in FIG. **2**A, in a retrieval position, distal end **204** of elongate member **202** may be distal of the distal end **104** of tube **102**. In an operating position, as shown in FIG. **2**B, distal end **204** of elongate member **202** may be closer to and/or flush with the distal end **104** of tube **102**.

(34) FIG. **3**A illustrates a cross-sectional view of an exemplary medical device **100**. This view may, for example, be at line **3-3** of FIG. **1**. As shown, tube **102** may include a lumen **112**, illumination device **126**, imaging apparatus **124**, and/or engaging surface **184**. Lumen **112** may be semi-circular in shape. Lumen **112**, however, may have any cross-sectional size and/or shape. In the example shown in FIG. **3**A, engaging surface **184** may be substantially planar. Engaging surface **184** is not limited thereto and may have shape, including any shape that is substantially complementary with an engaging surface of an elongate member (e.g., engaging surface **284** of tube **202**). For example, as shown in FIG. **3**A, elongate member **202** has a substantially planar engaging surface **284**. Lumen **214** of elongate member **202** may be substantially D-shaped, however, lumen **214** is not

limited thereto and may have any cross-sectional size and/or shape. FIG. **3**A shows a space between engaging surface **284** of elongate member **202** and engaging surface **184** of tube **102**. In some examples, however, gravity may pull and/or sleeve **302** may push engaging surface **284** of elongate member **202** to contact engaging surface **184** of tube **102**. Sleeve **302** should be configured to avoid so much friction as to restrict axial movement of elongate member **202** relative to tube **102**. Spaces **304** are spaces between an interior wall of sleeve **302** and the outer surfaces of elongate member **202** and/or tube **102**. In some examples, these spaces may be minimized, for example, if sleeve **302** is made of an elastic material configured to “hug” on the outer surfaces of elongate member **202** and/or tube **102**.

(35) FIG. **3**B illustrates a cross-sectional view of an alternative exemplary medical device **100**. This view may, for example, be at line **3-3** of FIG. **1**. As shown, the cross-sectional area of elongate member **202′** and lumen **214′** of elongate member **202′** are substantially circular. Tube **102** includes a lumen **112**, illumination device **126**, imaging apparatus **124**, and/or engaging surface **184′**. Engaging surface **184′** is substantially concave and may be substantially complementary to an outer surface of elongate member **202′**.

(36) It should be noted that in some examples, elongate member **202** and/or elongate member **202′** may be configured to engage (and/or slide across) an outer surface of a conventional or substantially circular delivery device. In such examples, engaging surface **284** and/or **284′** may be concave and/or be complementary with the outer surface of a substantially circular delivery device.

(37) B. Control System

(38) FIG. **4** is an enlarged exemplary view of proximal end **206** of elongate member **202** and control system **210**. Distal of proximal end **206**, elongate member **202** may fork, for example, into first end **214** and second end **244**. First end **244** of elongate member **202** may be open and configured to attach to, for example any fluid source known in the art, e.g., a fluid source **280** of FIG. **1**. For example, dye or, more specifically, radiopaque dye may be within fluid source **280**.

(39) Second end **214** of elongate member **202** may attach to control system **210** by, for example, a locking configuration, use of an adhesive, or integrally forming with elongate member **202**. Control system **210** may allow an operator to control operation of the retrieval system. For example, as described in further detail below, control system **210** may include the ability to control steering of elongate member **202**, introduce dye into elongate member **202**, and/or apply suction through elongate member **202**. In some examples, control system **210** may include or be in fluid communication with a vacuum source, e.g., vacuum source **250** of FIG. **1**. A vacuum source may include a house vacuum, vacuum pump, compressor unit, etc. Control system **210** may include any input devices known in the art, including buttons, switches, keyboards, dials, touchscreens, etc. Input device **230** may, for example, be an on-off button to turn on or off the vacuum source and either apply or cease application of suction from distal end **204** of elongate member **202**. Input device **230** may be a dial for adjusting the flow rate of this suction.

(40) IV. Insertion and Operation of the Medical Device

(41) FIG. **5**A-C illustrate an exemplary method of operation of medical device **100**, including an enlarged view of upper ureter **406** and lower calyx **410** of kidney **408**. Kidney stones **470** and **472** are located in lower calyx **410**. As previously mentioned, the space within calyx **410** of kidney **408** is often more limited than the space within the ureter **406**.

(42) An operator (e.g., a doctor or other medical professional) may connect elongate member **202** and tube **102** by sliding their respective distal ends through a proximal opening of sleeve **302**. In some examples, medical device **100** may be pre-assembled and this step may omitted. In some examples, one or both of tube **102** and elongate member **202** may be separately or simultaneously inserted into patient's urethra, through the urinary bladder, and into ureter **406**. In examples in which elongate member **202** is used as a guidewire, distal end **204** of elongate member **202** may be inserted into a patient's urethra, through the urinary bladder, and into ureter **406**. In some examples, once distal end **204** of elongate member **202** is positioned within the upper ureter, an operator may

initiate the introduction of fluid into kidney **408** through lumen **214** of elongate member **202**. For example, a fluid source (e.g., fluid source **280** of FIG. **1**) may be fluidly connected to first end **244** of elongate member **202** and fluid may be transmitted through first end **244**, through lumen **214** of elongate member **202**, to distal end **204** of elongate member **202**, and into the kidney **408**. In some examples, the fluid may be radiopaque dye. As shown in FIG. **5**A, dye **550** may enter the ureter **406** and/or kidney **408** (depending on the location of distal end **204** of elongate member **202**) and may disperse throughout kidney **408** and/or lower calyx **410**. This may allow the operator to determine the location of any kidney stones within the kidney. The introduction of any fluid (e.g., not limited to radiopaque) into the kidney may at least partially inflate the kidney, thus providing additional space to perform the desired procedure. Further, the introduction of fluid as an initial step may ensure that the pressure within the kidney remains at a sufficient level, e.g., the kidney will not collapse, once suction is applied to the kidney through lumen **214** of elongate member **202** (as described in further detail below). It should be noted that in some examples, fluid is not introduced through lumen **214** of elongate member **202**, and once disposed with the patient's body, an operator may proceed to the next step, e.g., locating and securing a kidney stone.

(43) Elongate member **202** may be extended distally into kidney **408**, so that the distal opening of lumen **214** is proximate to kidney stone **470**. At this point, elongate member **202** is in a retrieval position, e.g., distal end **204** of the elongate member **202** is positioned distal of distal end **104** of tube **102**. An operator may initiate a vacuum source, e.g., by selecting input device **230** of FIG. **4**, to apply suction through lumen **214** to distal end **204** of elongate member **202**. Suction into lumen **214** of elongate member **202** may pull the stone toward a distal opening of elongate member **202** until the stone contacts and forms a seal with the distal opening of elongate member **202**. The formed seal may secure the kidney stone to the distal opening of the elongate member **202**. As shown in FIG. **5**B, with the stone secured to elongate member **202**, elongate member **202** may be retracted or pulled proximally toward ureter **406** and/or distal end **104** of tube **102** until the elongate member is in the operating position, e.g., secured stone **470** is proximate a distal opening of tube **102**. At any time between the insertion of elongate member **202** into the patient and the above step of securing the stone proximate the distal opening of tube **102**, tube **102** may track elongate member **202** like a guidewire, so that distal end **104** is positioned within ureter **406**. FIG. **5**C illustrates an operating position of elongate member **202** in which distal end **204** of elongate member **202** is closer to distal end **104** of tube **102** than in the retrieval position.

(44) As shown in FIG. **5**C, once the laser fiber **120** is in a sufficient position to aim for secured kidney stone **470**, the operator may initiate the laser to break up kidney stone **470**. Laser fiber **120** may be introduced into a patient through lumen **112** of tube **102**. Laser fiber **120** may be connected to and/or controlled by laser control **130**. (It should be noted that the stone may be broken into particles in any way known in the art, including the application of ultrasound and/or sound waves on the stone.) Once stone **470** is broken into particles that are smaller than the cross-sectional area of the distal opening of elongate member **202**, the seal may be broken and/or particles may be suctioned into lumen **214** of elongate member **202**. These particles may be vacuumed through elongate member **202** and out of the patient's body. The stone particles are suctioned out from lumen **214** of elongate member **202** along with the used saline fluid and collect in the chamber outside the patient body through the saline outlet pathway.

(45) The application of suction through lumen **214** of elongate member **202** may improve the ability to break kidney stones by creating an anti-retropulsion effect. By applying suction through lumen **214**, a kidney stone may be pulled toward laser fiber **120**, thus countering the effect of the laser energy pushing the kidney stone away. This configuration thus assists in generating the smaller stone fragments by pulling the stones into the reach of laser fiber **120**.

(46) Once the operator determines a sufficient amount of the resulting particles of stone **470** have been removed from the patient's body or does not want to continue for other reasons, the laser process and/or application of suction may be stopped. At any point, an operator may additionally

choose to move the device within the patient. For example, an operator may choose repeat the process. In FIG. 5C, elongate member **202** has secured a first kidney stone **470**, but a second kidney stone **472** remains within lower calyx **410**. Once kidney stone **470** has been removed from the patient's body, the operator may move distal end **204** of elongate member **202** to a retrieval position and position its distal opening proximate to kidney stone **472**. The above process may then be repeated for securing, moving, and breaking up stone **472**. An operator may reposition any element of the medical device and/or repeat any of the above described steps any number of times. Once an operator determines no more kidney stones can and/or should be broken apart and/or no more stone fragments/dust can and/or should be removed, medical device **100** may be removed from the patient's body.

(47) FIG. **6** illustrates an exemplary method of use of a medical device for removal of stone fragments/dust. For purposes of discussion, method **600** will be described using medical device **100** of FIG. **1** and the urinary system of FIGS. 5A-C, as described above, but method **600** is not intended to be limited thereto. As shown in FIG. **6**, method **600** includes steps **602**, **604**, **606**, **608**, **610**, **612**, and **614**. However, it should be noted that method **600** may include more or fewer steps as desired for a particular implementation and the order of the steps may be varied.

(48) Method **600** may commence after elongate member **202** and/or tube **102** of medical device **100** have been inserted into a patient's body. In step **602**, distal end **204** of elongate member **202** may be positioned proximal of a target area (e.g., kidney **408** and/or lower calyx **410**). In step **604**, as illustrated in FIG. 5A, dye (e.g., dye **550**) may be supplied through a lumen (e.g., lumen **214**) of the elongate member (e.g., elongate member **202**) to the distal end (e.g., distal end **204**) of the elongate member. In step **606**, distal end **204** of elongate member **202** may be positioned at the target area (e.g., kidney **408** and/or lower calyx **410**) and/or distal of distal end **104** of tube **102**. Once distal end **204** of elongate member **202** is proximate an object (e.g., kidney stone **470**), method **600** may proceed to step **608** and suction may be applied through lumen **214** of elongate member **202**. An operator may apply suction to lumen **214** of elongate member **202** by, for example, initiating vacuum source **250** of FIG. **1**. The application of suction may move the object to the distal opening of lumen **214** and the object (e.g., kidney stone **470**) may form a seal with distal end **204** of elongate member **202**. In some examples, distal end **204** of elongate member **202** may be repositioned in order to capture the object with the applied suction. Once a seal is formed, method **600** may proceed to step **610**. In step **610**, distal end **204** of elongate member **202** may be moved distally and/or proximate to distal end **104** of tube **102**, as shown in FIG. 5B. Moving distal end **204** of elongate member **202** may also move the object (e.g., kidney stone **470**) distally and/or proximate to distal end **104** of tube **102**. In step **612**, once the object is proximate to distal end **104** of tube **102**, a laser fiber may be disposed within the tube. For example, laser fiber **120** may extend through lumen **112** to distal end **104** of tube **102** and may aim at kidney stone **470**. In step **614**, the laser may be initiated. For example, an operator may initiate the laser through laser control **130** and/or handle portion **110** of FIG. **1**.

(49) The many features of the disclosure are apparent from the detailed specification, and thus, it is intended by the appended claims to cover all such features of the disclosure which fall within the true spirit and scope of the disclosure. Further, since numerous modifications and variations will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation illustrated and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure.

(50) Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

1. A medical device, comprising: a delivery device including a tube having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end; and a member having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end, wherein the lumen is configured to apply suction through the distal end of the member, wherein at least a distal portion of the member is D-shaped with a flat side and a rounded side extending from opposing ends of the flat side; wherein at least one of the tube or the member is axially movable relative to the other of the tube or the member; wherein at least a portion of a circumferential exterior of the tube abuts a portion of the flat side of the member; and wherein an outer portion of the tube includes a flat engaging surface that is complementary to the flat side of the distal portion of the member.
2. The medical device of claim 1, wherein the delivery device is one of a ureteroscope, a sheath, a catheter, or an endoscope.
3. The medical device of claim 1, wherein the distal end of the tube includes an illumination device and an imaging device, and wherein the medical device further comprises a laser fiber connected to the tube.
4. The medical device of claim 1, wherein the member is movable to extend distally beyond the tube to apply suction to an object at a first position and retract proximally to bring the object to a second position closer to the distal end of the tube.
5. The medical device of claim 1 wherein the tube includes a partially circular portion extended from opposite sides of the flat engaging surface.
6. The medical device of claim 1, further comprising a sleeve, wherein the delivery device and the member are adjacent to each other at least partially within the sleeve.
7. A medical device, comprising: a tube having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end; and a member having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end, wherein the lumen is configured to apply suction through the distal end of the member, wherein at least a distal portion of the member is D-shaped with a flat side and a rounded side extending from opposing ends of the flat side, wherein at least a portion of a circumferential exterior of the tube includes a flat engaging surface that is complementary to and abuts a portion of the flat side of the member, wherein the tube includes a partially circular portion extended from opposite sides of the flat engaging surface, and wherein at least one of the tube or the member is axially movable relative to the other of the tube or the member.
8. The medical device of claim 7, further comprising a sleeve, wherein the tube and the member are adjacent to each other at least partially within the sleeve.
9. The medical device of claim 8, wherein the distal end of the tube includes an illumination device and an imaging device, and wherein the medical device further comprises a laser fiber connected to the tube.
10. The medical device of claim 7, wherein the member is movable relative to the tube to extend distally beyond the tube to apply suction to an object at a first position and retract proximally to bring the object to a second position closer to the distal end of the tube.
11. The medical device of claim 10, further comprising a laser element within the lumen of the tube, wherein the laser element is longitudinally moveable within the lumen of the tube to extend distally beyond the distal end of the tube.
12. The medical device of claim 7, wherein the distal end of the member includes a reinforced section.
13. The medical device of claim 7, wherein the medical device includes a handle, wherein the member is operably coupled to a control system, wherein the tube is operably coupled to the

handle, wherein the member is configured to be coupled to a suction source, and wherein the member is movable relative to the tube to extend distally beyond the tube to apply suction to an object at a first position and retract proximally to bring the object to a second position closer to a distal end of the tube.

14. A medical device, comprising: a delivery device including a tube having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end; and a member having a distal end, a proximal end, and a lumen extending between the proximal end and the distal end, wherein the lumen is configured to apply suction through the distal end of the member, wherein at least a distal portion of the member is D-shaped with a flat side and a rounded side extending from opposing ends of the flat side; wherein the member is axially movable relative to the tube such that the member is extendable distally of the tube, wherein at least a portion of a circumferential exterior of the tube abuts a portion of the flat side of the member, wherein an outer portion of the tube includes a flat engaging surface that is complementary to the flat side of the distal portion of the member, and wherein another outer portion of the tube includes a rounded portion extending from opposite sides of the flat engaging surface.

15. The medical device of claim 14, wherein the delivery device is one of a ureteroscope, a sheath, a catheter, or an endoscope.

16. The medical device of claim 14, wherein the distal end of the tube includes an illumination device and an imaging device, and wherein the medical device further comprises a laser fiber connected to the tube.

17. The medical device of claim 14, wherein the member is movable to extend distally beyond the tube to apply suction to an object at a first position and retract proximally to bring the object to a second position closer to the distal end of the tube.

18. The medical device of claim 14, wherein the tube includes a partially circular portion extended from opposite sides of the flat engaging surface.

19. The medical device of claim 14, further comprising a sleeve, wherein the delivery device and the member are adjacent to each other at least partially within the sleeve.

20. The medical device of claim 19, wherein the medical device includes a handle, wherein the member is operably coupled to a control system, wherein the tube is operably coupled to the handle, wherein the member is configured to be coupled to a suction source, and wherein the member is movable relative to the tube to extend distally beyond the tube to apply suction to an object at a first position and retract proximally to bring the object to a second position closer to a distal end of the tube.